# SRI VENKATESWARA COLLEGE OF ENGINEERING (AUTONOMOUS)

## DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

**SECTION-E** **Project ID-EC26E11**

# Raspberry Pi-Enabled Multi-Modal AI System for Deepfake Detection in Multimedia


**TEAM MEMBERS:**

1. P.Pavan Kalyan Reddy (23BFA04L34)
2. P.Sravanthi (22BFA04317)
3. T.Baby Shalini (22BFA04324)
4. N.Bhavya (22BFA04298)
5. D.Ajay (23BFA04L20)

**UNDER THE ESTEEMED GUIDANCE OF:**

Dr.E.Anant Shankar, Ph.D.,
Associate Professor

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# PROBLEM IDENTIFICATION

- Deepfake is a technology that creates fake images, videos, or audio that look real.
- Deepfakes can cause misinformation, fraud, and identity misuse
- Most existing detection systems focus on a single modality (image or video only).
- Cloud-based systems are slow and fully depend on internet connectivity
- There is a need for a fast, multi-modal, and portable detection system

# MOTIVATION OF THE PROJECT

- Need for a simple and affordable deepfake detection system
- Requirement for detecting multiple types of media (image, video, and audio)
- Interest in implementing AI models on embedded systems
- Aim to achieve efficient detection using Raspberry Pi
- Development of a portable and user-friendly solution

# TITLE JUSTIFICATION

- **Raspberry Pi-Enabled** → Low-cost and portable embedded platform
- **Multi-Modal** → Supports image, video, and audio inputs
- **AI System** → Uses deep learning models for detection
- **Deepfake Detection** → Identifies fake and manipulated media
- **Multimedia** → Works on different types of digital content

# WHAT WORK MODIFIED IN BASE PAPER

- The base paper focused only on video-based deepfake detection
- Our project enhances this by:

1. Extending to multi-modal detection (images, video, audio)
2. Using MobileNetV2 for image and video classification
3. Using LCNN model for audio deepfake detection
4. Deploying the system on Raspberry Pi using TensorFlow Lite

# OBJECTIVES

- Detect deepfake content in images, videos, and audio using deep learning techniques.
- Develop a multi-modal system that can handle different types of media inputs.
- Design an efficient classification approach for identifying real and fake content.
- Implement the system on Raspberry Pi for portable and edge-based processing.
- Display the detection results on an LCD and provide a buzzer alert for fake content.

# ABSTRACT


- Deepfakes are AI-generated multimedia content (images, videos, and audio) that appear realistic and are difficult to detect.
- This project presents a deepfake detection system using Raspberry Pi to analyze live and stored media inputs.
- MobileNetV2 is used for image and video detection, and LCNN is used for audio analysis.
- Preprocessing techniques such as resizing, frame extraction, and spectrogram generation are applied before model inference.
- The models are optimized using TensorFlow Lite for efficient edge deployment.
- The system displays results on LCD and activates a buzzer alert when fake content is detected.

# INTRODUCTION

- Deepfakes are AI-generated fake images, videos, and audio that look real.
- They are widely used in cybercrime, fake news, and digital manipulation.
- This project develops a multi-modal deepfake detection system using Raspberry Pi.
- MobileNetV2 is used for image and video analysis, and LCNN is used for audio detection.
- The system processes input media and classifies it as real or fake using optimized deep learning models.

# Existing Method

- Most existing systems use complex deep learning models, requiring high computational power and GPUs.
- Many approaches rely on heavy architectures, making them unsuitable for real-time and low-resource environments.
- Several methods are cloud-based, causing latency, internet dependency, and privacy concerns.
- Many systems depend on single-modality data, limiting robustness in detection.
- Processing full video data leads to high latency and difficulty in edge-device deployment.

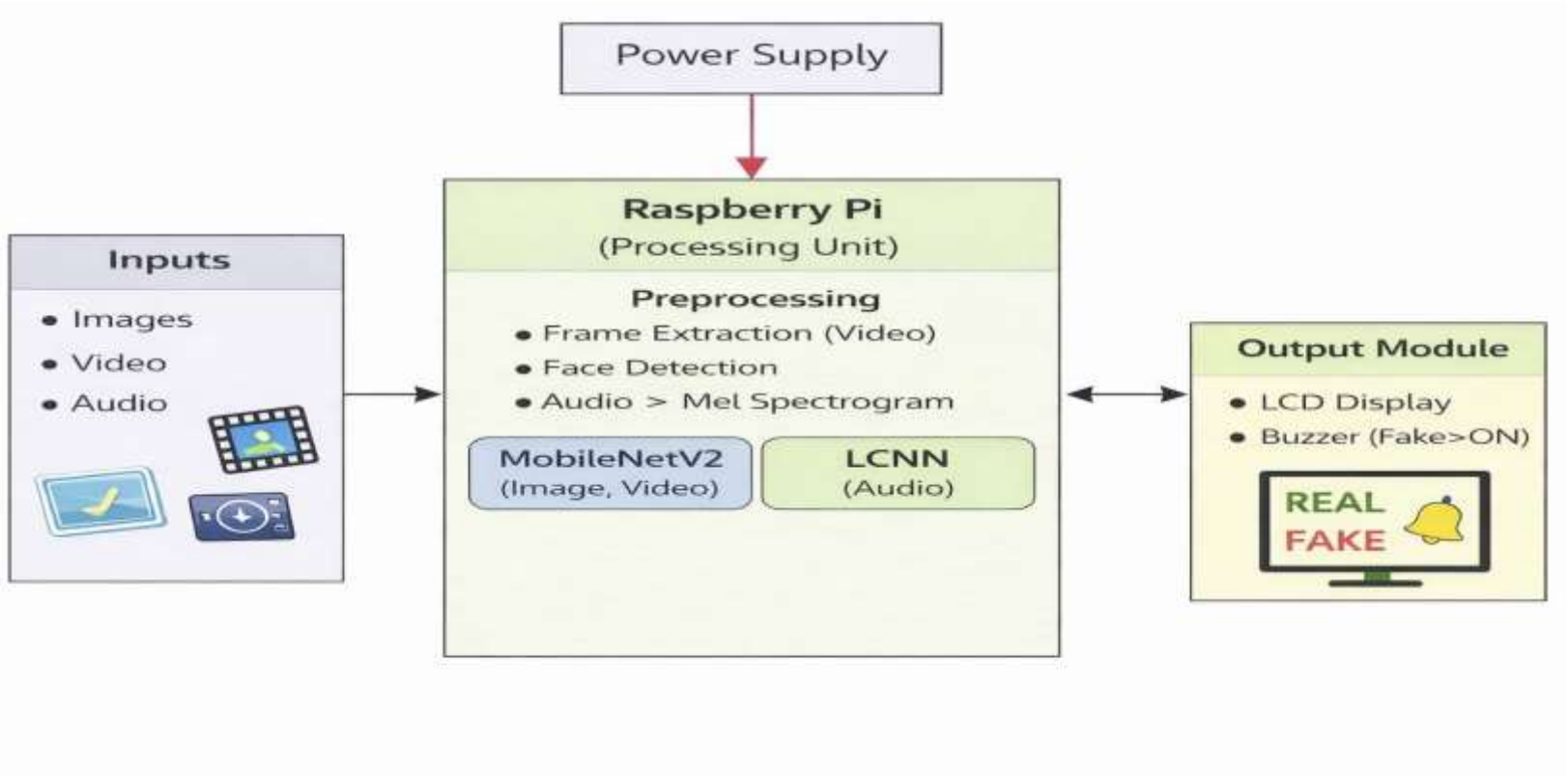
# Dis- Advantages

1. Lower accuracy in detecting high-quality and advanced deepfakes.
2. Requires high computational power (GPU dependency).
3. Slow processing speed, causing delays in detection.
4. Difficult to deploy on edge or low-resource devices.
5. Limited feature extraction capability, missing subtle manipulations.
6. Many systems are cloud-dependent, leading to latency and privacy concerns.

# Proposed Method

- The system performs multimodal deepfake detection using image, video, and audio inputs.
- MobileNetV2 is used to analyze images and extracted video frames.
- LCNN is used to analyze audio using mel spectrograms.
- Video input is processed by frame extraction, and each frame is analyzed individually.
- Each input (image, video, audio) is evaluated independently and assigned a prediction score.
- Results are displayed on LCD, and a buzzer is activated for fake detection.
- A camera interface is used for testing only, while the system mainly works on pre-recorded inputs.

# Block Diagram



# METHODOLOGY

## 1. Data Acquisition

- The system accepts multimedia input from the user
- Input can be:
    1. Image file
    2. Video file
    3. Audio file
- The user provides the file path through the Python interface
- The system loads the selected media into Raspberry Pi memory

# METHODOLOGY

## 2. Preprocessing

- Image Input:
    1. Resize to 224 × 224
    2. Normalize pixel values
- Video Input:
    1. Extract frames using OpenCV
    2. Resize each frame to 224 × 224
- Audio Input:
    1. Convert audio to Mel spectrogram
    2. Resize spectrogram to 224 × 224

# METHODOLOGY

## 3. Feature Extraction

- Extract meaningful patterns from input media

Image and Video Features:

1. Texture inconsistencies
2. Facial distortions
3. Pixel-level artifacts

Audio Features:

1. Frequency variations
2. Speech pattern irregularities

# METHODOLOGY

## 4. Model Processing

- MobileNetV2 is used for image and video analysis
- LCNN (Light CNN) is used for audio analysis
- Models are converted to TensorFlow Lite format
- Deployed on Raspberry Pi for efficient processing
- Each model predicts whether input is real or fake

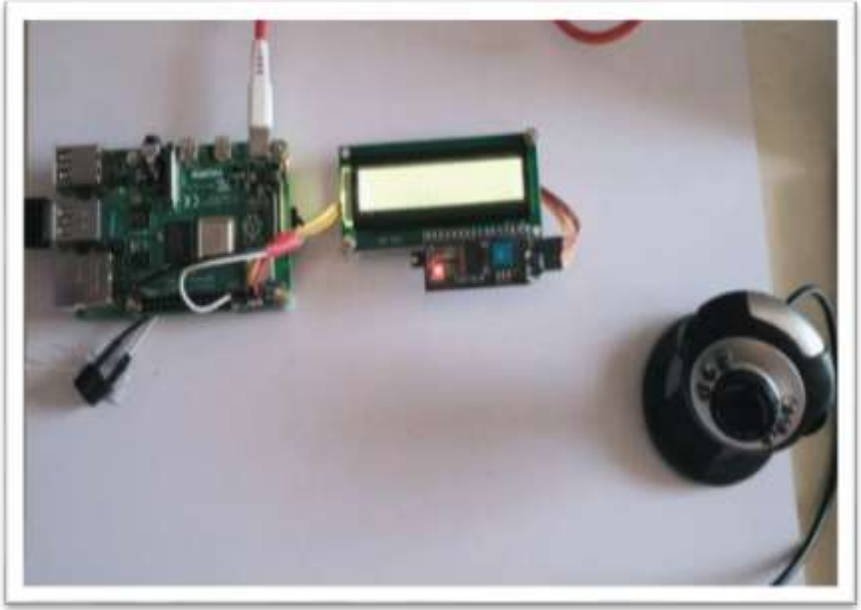
# METHODOLOGY

## 5. Classification And Output

- The system classifies media as:
    1. REAL
    2. FAKE
- Output is displayed on:
    1. Monitor (Python output)
    2. LCD display
- If fake content is detected then Buzzer is activated

# HARDWARE DESCRIPTION

- Raspberry Pi 4 Model B
- USB Camera
- Micro SD Card
- Power Supply (5V Adapter)
- 16x2 I2C LCD Display
- Buzzer



# HARDWARE DESCRIPTION

1. Raspberry Pi 4 Model B

- Acts as the main processing unit for running deep learning models and controlling the system
- Handles input processing, model inference, and output display

2. USB Camera

- Captures real-time images and video for deepfake detection
- Provides input directly to the Raspberry Pi4.

3. MicroSD Card

- Stores the operating system, trained models, and project files
- Acts as the primary storage for the Raspberry Pi

# HARDWARE DESCRIPTION

4. Power Supply

- Provides required power (5V, 3A) to operate the Raspberry Pi and connected components

5. 16×2 I2C LCD

- Displays the detection result (REAL / FAKE)
- Uses I2C interface to reduce wiring complexity

6. Buzzer

- Generates an alert sound when fake media is detected
- Helps in quick user notification

7. Laptop / PC (Support System)

- Used for uploading code and providing input media paths
- Acts as a monitoring interface during testing and execution

# SOFTWARE DESCRIPTION

- Raspberry Pi OS — Operating system used to run the system on Raspberry Pi
- Python — Main programming language for system implementation
- OpenCV — Used for image processing, face detection, and video frame extraction
- TensorFlow Lite — Used for running trained models efficiently on Raspberry Pi
- Keras — Used for designing and training deep learning models
- NumPy — Used for numerical operations and data preprocessing
- Librosa — Used for audio processing and Mel spectrogram generation
- VNC Viewer — Used for remote access and control of Raspberry Pi
- Advanced IP Scanner — Used to find Raspberry Pi IP address in network

# Advantages

1. Supports multiple media types (image, video, and audio)
2. Runs on Raspberry Pi without requiring high-end hardware
3. Portable and compact system for easy deployment
4. Low-cost implementation using affordable components
5. Works with both live camera input and stored media files
6. Uses lightweight deep learning models for faster processing
7. Provides clear output through LCD display and buzzer alert
8. Simple and user-friendly system for basic operation

# Applications

1. Detect fake images, videos, and audio before sharing on social media
2. Verify authenticity in online meetings, interviews, and exams
3. Assist digital forensic investigations in identifying manipulated media
4. Help journalists verify news content before publishing
5. Used in surveillance systems to detect fake multimedia content
6. Useful for education and research in deepfake detection

# Results

- The system accepts image, video, or audio input from the user
- The model processes the input and classifies it as Real or Fake
- The result is displayed on the monitor with prediction output
- The same result is shown on the LCD display
- A buzzer alert is activated when fake media is detected
- The system works on Raspberry Pi using TensorFlow Lite models

# Results

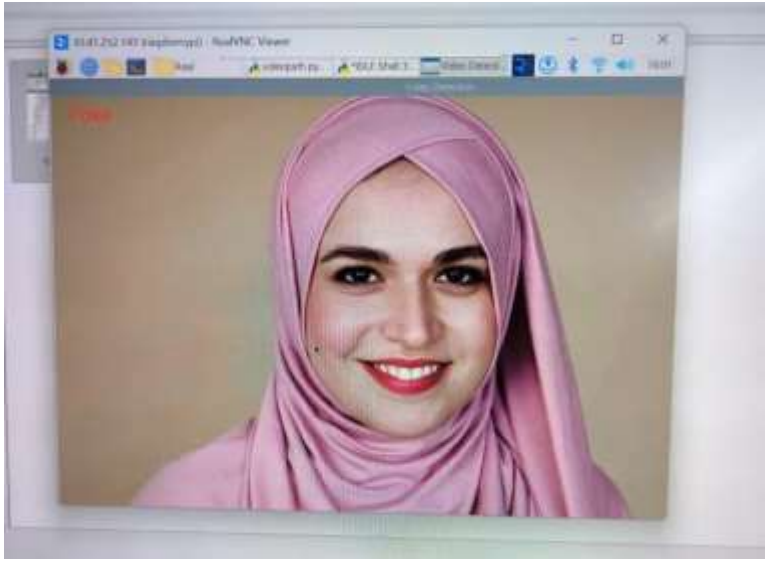


Figure 1: Model Output

- Input image is analyzed and classified as Fake by the system
- Output is displayed on the monitor

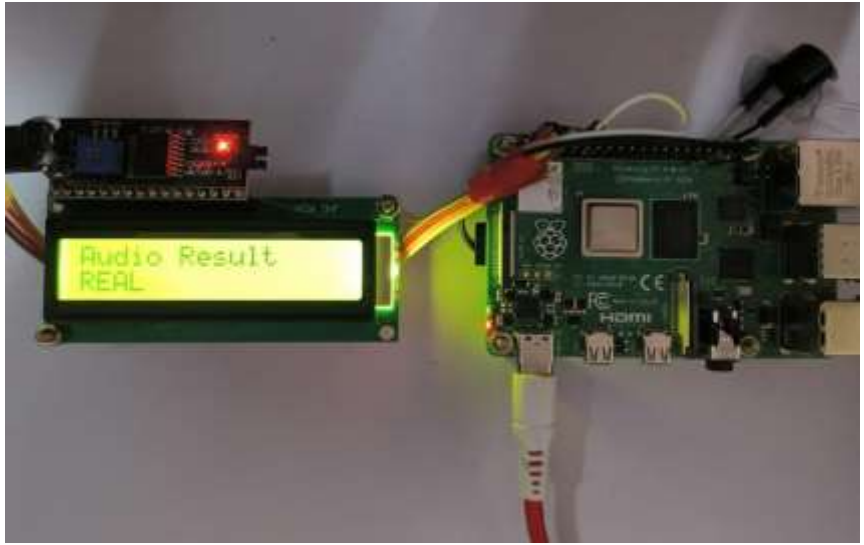



Figure 2: Hardware Output

- Detection result is shown on LCD display
- Confirms working of Raspberry Pi-based system

# CONCLUSION

- Developed a multi-modal deepfake detection system for image, video, and audio
- Implemented on Raspberry Pi using lightweight deep learning models
- MobileNetV2 (image,video) and LCNN (audio) used for efficient detection
- Performs preprocessing and analysis to identify manipulated media
- Provides output through monitor, LCD display, and buzzer alert
- Designed as a low-cost, portable, and easy-to-use solution
- Suitable for practical applications like media verification and security

# FUTURE SCOPE

- Improve model accuracy using larger and more diverse datasets
- Extend system to support real-time streaming inputs
- Enhance audio deepfake detection using advanced models
- Optimize model performance for faster execution on Raspberry Pi
- Improve system reliability under different lighting conditions
- Develop a user-friendly interface for easier operation

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